Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2024	ENGI3371-WE01

Title: (BEng/MEng)

Electrical Engineering 3

Time Allowed:	2 hours
Additional Material provided:	Yes, formula sheet included as part of exam paper.
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	This is a synchronous, time restricted exam. All submissions must be uploaded to Gradescope as per the published timetable. Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the exam submission window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Section A

Question 1

a) Sketch the circuit diagram of a 2Q bi-directional DC-DC converter to feed a separately excited DC machine load. With the aid of a circuit diagram, briefly explain the buck and boost modes of operation of this converter.

[30%]

b) A single-phase full-wave thyristor controlled rectifier is used to control the speed of a 15 hp, 150 V, 1200 rpm, separately excited DC motor. The rectifier is connected to a single-phase power supply with 230 V, 50 Hz. The armature resistance is $R_a = 0.6\,\Omega$ and armature inductance is $L_a = 7.5\,mH$. The motor voltage constant is $k\phi = 0.12\,V/rpm$. This DC motor is considered as a highly inductive load, so the output voltage of the rectifier is given by:

$$V_{DC} = \frac{2V_m}{\pi}\cos\alpha$$

where V_m is the peak voltage of the AC source.

This motor is operating at 1000 rpm, and carries a ripple-free armature current of 36 A.

i) Sketch a schematic diagram of this system and clearly label all components.
ii) Determine the internal voltage of the motor E_a, and the terminal voltage V_a.
iii) Determine the firing angle of the thyristor α.
iv) Calculate the electrical power delivered to the motor.

[70%]

continued

Question 2

a) For a three-phase induction motor, sketch a block diagram of a scalar control system and clearly label all components. Briefly explain how this control system can be used for dynamic electromagnetic breaking, and name one industrial application.

[30%]

b) Consider a four pole three-phase induction motor carrying a constant mechanical load at its mechanical shaft. This motor is initially supplied with rating voltage and frequency of 400 V, 50 Hz.

i) Aided by sketches, qualitatively describe the change of operating point on the torque-speed characteristics if the supply is changed to 200 V, 25 Hz, of the same phase sequence.

ii) Considering a constant rotor slip of $S = 5\ \%$, calculate synchronous speed n_s and rotor speed n_r in rpm for both operating points.

[20%]

c) A Y-connected three phase synchronous generator is connected to a 6.6 kV infinite bus. It has a synchronous reactance of 1.8 Ω per phase, and the armature resistance is negligible. The generator delivers an active power output of 26 MW and a reactive power output of 12 MVAR to the infinite bus.

i) Sketch a per-phase circuit diagram of the generator and label all components.

ii) Determine phasor of the armature current $\boldsymbol{I}_A$ and internal voltage of the generator $\boldsymbol{E}_A$.

iii) Plot the phasor diagram including the terminal voltage, excitation voltage, armature current and voltage drop across the synchronous reactance.

[50%]

continued

Section B

Question 3

a) Show, using principles of circuit theory (KVL and KCL), that the generalised circuit constants for the per-phase medium-length transmission line model (modelled as the nominal-π circuit) are defined by equations below:

$$A = D = \frac{ZY}{2} + 1$$

$$B = Z \ \ \Omega$$

$$C = Y\left(1 + \frac{ZY}{4}\right) \ \ S$$

in which Z is the total (lumped) per-phase series impedance of the line and Y is the total (lumped) per-phase shunt admittance of the line. Meanwhile, draw the per-phase nominal-π circuit model of the medium-length transmission line.

[30%]

b) A 50Hz, three-phase transmission line is 200 km long and has the following parameters per phase per unit-length:

Table Q3 – Transmission Line Parameters (per phase per unit length)

Series Resistance (r)	Series Reactance (x)	Shunt Admittance (y)
$0.0275 \ \ \Omega/km$	$0.16\,\Omega/km$	$2.16 \times 10^{-6}\angle 90°\ S/km$

The line delivers 400 MVA at 345 kV (line-line, RMS) to a star-connected load at 0.8 lagging power factor.

Calculate the following:

i) The active (real) and reactive powers supplied to the load at the receiving end.

ii) The magnitude and phase angle of the current supplied to the load at the receiving end.

iii) The magnitude and phase angle of the sending-end voltage (line-line) using (1) the per-phase short-length transmission line model and (2) the per-phase medium-length transmission line model.

iv) Is the medium-length transmission line model a good model for this scenario? Explain your answer.

[70%]

continued

Question 4

a) A thermal power plant has two generating units with the following cost functions:

$$C_1 = 0.004P_1^2 + 8P_1 + 500 \quad £/h$$

$$C_2 = 0.0048P_2^2 + 6.4P_2 + 400 \ £/h$$

The plant is used to supply 250 MW of demand. The minimum power limit for each generator is zero MW.

Answer the following questions:

i) Compute the plant's marginal cost at which the optimal (most economic) division of load between the two generators is achieved.
ii) Compute the total cost of supplying the load under these conditions.
iii) Would it be feasible to economically divide the load between both generating units if the load suddenly dropped to 80 MW? Justify your answer.

[30%]

b) A power system has three buses and three lines. A single load is connected to Bus 2 with a total value of $170 + j100$ MVA. A generator is connected to Bus 3 with a total generation of 50 MW. Bus 1 is the slack bus and the voltage magnitude in buses 1 and 3 are both at 1.05 p.u. Table Q4 shows the line series impedance data for this power system in per unit on a 100 MVA base.

Table Q4 – Line Series Impedance Data for the 3-bus Power System

Line Number	Connections	Line Series Impedance (p.u.)
Line 1	Bus 1 to Bus 2	$0.02 + j0.04$
Line 2	Bus 1 to Bus 3	$0.01 + j0.03$
Line 3	Bus 2 to Bus 3	$0.0125 + j0.025$

Draw the single line diagram of this system clearly labelling the line impedance values as well as the values of load and generation at each bus (if applicable).

Determine the following:

i) The network admittance matrix (Y_{bus}) and the B' and B'' matrices.
ii) The power flow solution by performing one iteration of the fast-decoupled method.

[70%]

continued

Formula sheet:

$$\Delta\delta = -[B']^{-1}\frac{\Delta P}{|V|}$$
$$\Delta|\mathrm{V}| = -[\mathrm{B}'']^{-1}\frac{\Delta Q}{|V|}$$

B' and B'' are fixed matrices and can be obtained from the imaginary part of the bus admittance matrix Y_{bus}.

$$P_i = \sum_{j=1}^{n}|V_i||V_j||Y_{ij}|\cos(\theta_{ij} - \delta_i + \delta_j)$$
$$Q_i = -\sum_{j=1}^{n}|V_i||V_j||Y_{ij}|\sin(\theta_{ij} - \delta_i + \delta_j)$$

END